

0 2

Dengue is a serious disease that is caused by a virus. The virus is carried from one person to another by a mosquito, *Aedes aegypti*. One method used to try to reduce transmission of this disease is the Sterile Insect Technique (SIT). This involves releasing large numbers of sterile (infertile) male *A. aegypti* into the habitat. These males have been made infertile by using radiation.

0 2 . 1

Explain how using the SIT could reduce transmission of dengue.

[2 marks]

0 2 . 2

Describe how the mark-release-recapture method could be used to determine the population of *A. aegypti* at the start of the investigation.

[3 marks]

Question 2 continues on the next page

0 2 . 3

The release of radiation-sterilised *A. aegypti* has not been very successful in controlling the transmission of dengue.

Suggest **one** reason why.

[1 mark]

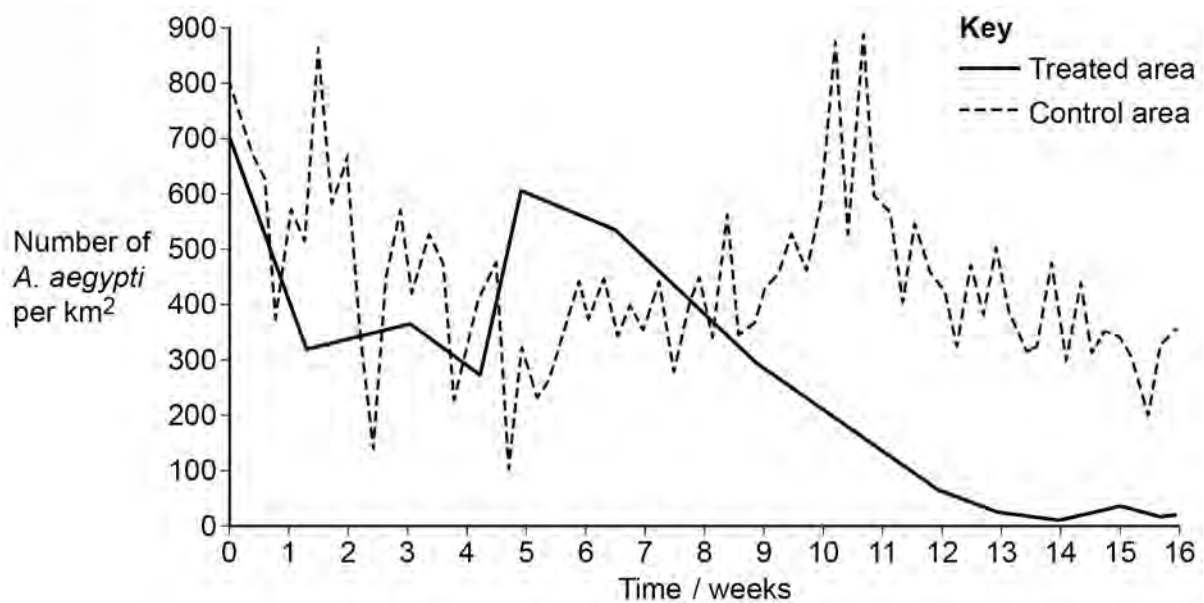
0 2 . 4

Recently a new method was developed to control *A. aegypti*. Scientists produced transgenic males carrying a 'lethal gene' which kills their offspring before they can reproduce.

The scientists released transgenic males every week in one area of a city in Brazil. At regular intervals they determined the number of *A. aegypti* per km² in the area where transgenic males were released and in a control area where no transgenic males were released.

Figure 2 shows their results.

Figure 2



Suggest why the scientists released more transgenic males every week.

[1 mark]

0 2 . 5

The release of transgenic males proved successful in reducing the number of *A. aegypti*.

Describe how the results in **Figure 2** support this conclusion.

[2 marks]

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